Rajalakshmi Engineering College

Department of Computer Science and Design

GE23627 DESIGN THINKING AND INNOVATION

AY 2026–2027

S.C.O.P.E.S – Team Scope Sheet

Scoping of design challenge: A design challenge begins with a specific, purposeful situation or problem to solve. It should be clear, actionable and achievable. The problem statement describes the desired outcome rather than presenting a solution. SCOPES is used to frame and explore the challenge.

TEAM NAME	**OmniDB**
TEAM MEMBERS	1. Balamurugan RMM (241701008) 2. Vaishnav Perumal M (241701061)
DOMAIN NAME	Cross-Platform Gaming Information and Price Comparison System
DATE	7.8.2026
FACULTY INCHARGE	Dr. P. Revathy

S – Situation / Problem • What is the real situation you want to address? • Where does this problem occur? • Describe without mentioning a solution.	1. Game information is spread across separate storefronts and platform-specific websites, making it difficult for players to view games across PC, PlayStation, Xbox and Nintendo ecosystems in one place. 2. Players may need to visit multiple sources to compare editions, DLCs, developers, supported platforms and prices. 3. Tracking wishlisted games, owned games, price changes and total spending can be inconvenient when information is scattered across services.

C – Constraints What are the CONSTRAINTS when trying to address the situation or problem?	• Game metadata and availability can differ between platforms and regions. • Prices and discounts change frequently and may have different currencies or editions. • Accurate information depends on reliable, permitted data sources; scraping must respect site terms and access restrictions. • The catalogue must avoid duplicate entries for the same game while preserving platform-specific editions. • A growing catalogue requires consistent data structure and validation. • User library, wishlist and spending records require appropriate privacy and data protection. • A student project has limited development time, resources and access to official APIs.

O – Objectives & Outcomes What are the OBJECTIVES and intended OUTCOME of this design challenge?	**Objectives** 1. To provide a unified catalogue of games across major gaming platforms and storefronts. 2. To make game discovery easier through search, filters and structured game details. 3. To help users compare available platforms, editions and prices where data is available. 4. To let users organize a wishlist and personal game library. 5. To help users record game purchases and understand their total spending through a virtual wallet tracker. **Outcomes** 1. Less time spent switching between multiple game information sources. 2. Easier comparison of game availability and pricing. 3. A more organized view of games users own or want to buy. 4. Better awareness of recorded gaming expenditure. 5. A scalable foundation for expanding the catalogue and adding verified data sources.

P – People Who are the PEOPLE that you are trying to help? Whose problem are you trying to address? **Primary Stakeholders** Secondary Stakeholders Decision Makers	**Primary Stakeholders** a) PC and console gamers b) Players comparing games across platforms c) Users managing a wishlist, library and game purchases **Secondary Stakeholders** a) Game developers and publishers whose titles are listed b) Digital storefronts and platform providers c) Gaming communities and reviewers who use game information **Decision Makers** a) Project team members b) Faculty guide / project evaluator c) Future system administrator or maintainer

E – Estimates What are the ESTIMATED resources required for this design challenge?	**Human Resources** • Student project team • Faculty guide / project reviewer **Software / Technology** • Google AI Studio – existing frontend development • Backend – planned, not yet implemented • Web browser and code editor / development environment • Database and API integration to be finalized as backend development progresses **Hardware** • Laptop or desktop computer • Smartphone for responsive testing • Internet connection **Data Resources** • Verified game metadata and platform information • Permitted official APIs or other legitimate data sources **Time** • Approximately 4–6 weeks

S – Scope to Explore Is there ample SCOPE to explore multiple alternatives to address the problem?	There is scope to explore multiple alternatives for improving cross-platform game discovery and purchase tracking, such as: • A unified game catalogue covering PC, PlayStation, Xbox, GOG and Nintendo platforms. • Search, filters and sorting by platform, genre, developer and other available metadata. • Game pages with editions, DLCs, developers and supported platforms. • Price comparison where reliable data is available. • Wishlist and personal library management. • Virtual wallet / purchase tracker to record spending on games. • Duplicate detection and data validation for catalogue quality. • Responsive interface for desktop and mobile use. • Future backend and database integration for persistent user data and catalogue updates.

Frame the Design Challenge Write a clear problem statement that describes the desired outcome, not the proposed solution.	Gamers who use multiple platforms and digital storefronts often find it difficult to access consistent game information, compare editions and prices, and keep track of games they own, want to buy and have spent money on. Information is distributed across different services, requiring users to switch between sources and maintain purchase records separately. How might we make cross-platform game discovery, comparison and personal game tracking more convenient and organized for gamers?

SIGNATURE